EXERCISE #1

1)

INPUT: POLICY π, MDP

OUTPUT: V_π (AN ESTIMATION OF)

- INITIALIZE $V(s)$ $\forall s \in S$ ARBITRARILY
  (EXCEPT $V(\text{terminal}) = 0$)

- LOOP FOR $i = 1:K$

  LOOP FOR EACH $s \in S$

  $$V(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)\left[r + \gamma V(s')\right]$$

  END

END

2) START: $V = (0, 0, 0, 0)$ WE WILL NOT UPDATE THE TERMINAL STATE

$$V(s_1) \leftarrow \frac{1}{3}\overset{a_1}{\left(0 + V(s_1)\right)} + \frac{1}{3}\overset{a_2}{\left(-3 + V(s_2)\right)} + \frac{1}{3}\overset{a_3}{\left(-9 + V(s_3)\right)}$$

$$V(s_2) \leftarrow \frac{1}{3}\overset{a_1}{\left(-3 + V(s_2)\right)} + \frac{1}{3}\overset{a_2}{\left(-6 + \underset{0}{V(s_4)}\right)} + \frac{1}{3}\overset{a_3}{\left(-3 + V(s_3)\right)}$$

ALWAYS ZERO!

$$V(s_3) \leftarrow \frac{1}{3}\overset{a_1}{\left(0 + V(s_3)\right)} + \frac{1}{3}\overset{a_2}{\left(-2 + \underset{0}{V(s_4)}\right)} + \frac{1}{3}\overset{a_3}{\left(0 + V(s_2)\right)}$$

ALWAYS ZERO!

$\boxed{k = 1}$

$V(s_1) \Leftarrow \frac{1}{3}\left(-12 + \underset{0}{V(s_1)} + \underset{0}{V(s_2)} + \underset{0}{V(s_3)}\right) = -4$ $\quad V = (-4, 0, 0, 0)$

$V(s_2) \Leftarrow \frac{1}{3}\left(-12 + \underset{0}{V(s_2)} + \underset{0}{V(s_3)} + \underset{0}{V(s_4)}\right) = -4$ $\quad V = (-4, -4, 0, 0)$

$V(s_3) \Leftarrow \frac{1}{3}\left(-2 + \underset{-4}{V(s_1)} + \underset{0}{V(s_3)} + \underset{0}{V(s_4)}\right) = -2$ $\quad V = (-4, -4, -2, 0)$

$\boxed{k=2}$

$$V(s_1) \Leftarrow \frac{1}{3}\Big(-12 + \underset{-4}{V(s_1)} + \underset{-4}{V(s_2)} + \underset{-2}{V(s_3)}\Big) = -\frac{22}{3}$$
$V = (-\frac{22}{3}, -4, -2, 0)$

$$V(s_2) \Leftarrow \frac{1}{3}\Big(-12 + \underset{-4}{V(s_2)} + \underset{-2}{V(s_3)} + \underset{0}{V(s_4)}\Big) = -6$$
$V = (-\frac{22}{3}, -6, -2, 0)$

$$V(s_3) \Leftarrow \frac{1}{3}\Big(-2 + \underset{-6}{V(s_2)} + \underset{-2}{V(s_3)} + \underset{0}{V(s_4)}\Big) = -\frac{10}{3}$$
$V = (-\frac{22}{3}, -6, -\frac{10}{3}, 0)$

$\boxed{k=3}$

$$V(s_1) \Leftarrow \frac{1}{3}\Big(-12 + \underset{-\frac{22}{3}}{V(s_1)} + \underset{-6}{V(s_2)} + \underset{-\frac{10}{3}}{V(s_3)}\Big) = \frac{1}{9}\left(-36-22-18-10\right)$$
$$= -\frac{86}{9}$$
$V = (-\frac{86}{9}, -6, -\frac{10}{3}, 0)$

$$V(s_2) \Leftarrow \frac{1}{3}\Big(-12 + \underset{-6}{V(s_2)} + \underset{-\frac{10}{3}}{V(s_3)} + \underset{0}{V(s_4)}\Big) = \frac{1}{9}\left(-36-18-10\right) = -\frac{64}{9}$$
$V = (-\frac{86}{9}, -\frac{64}{9}, -\frac{10}{3}, 0)$

$$V(s_3) \Leftarrow \frac{1}{3}\Big(-2 + \underset{-\frac{64}{9}}{V(s_2)} + \underset{-\frac{10}{3}}{V(s_3)} + \underset{0}{V(s_4)}\Big) = \frac{1}{27}\left(-18-64-30\right) = -\frac{112}{27}$$
$V = (-\frac{86}{9}, -\frac{64}{9}, -\frac{10}{3}, 0)$

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3)

INPUT: POLICY π, MDP
OUTPUT: POLICY π'
- INITIALIZE $V(s)$ $\forall s \in S$ ARBITRARILY (EXCEPT $V(\text{terminal}) = 0$)
- POLICY EVALUATION (AS IN ANSWER TO POINT 1)
- POLICY IMPROVEMENT
  WHILE 'POLICY π' IS NOT STABLE
    FOR EACH $s \in S$
      $\pi(s) \leftarrow \arg\max_a \sum_{s',r} p(s',r|s,a)\left[r + \gamma V(s')\right]$
    END
  END

4) $V_\pi = \left(-13, -\frac{26}{3}, -\frac{16}{3}, 0\right)$

$$\pi'(s_1) = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: & 0 + V(s_1)_{-13} \\ a_2: & -3 + V(s_2)_{-\frac{26}{3}} \\ a_3: & -9 + V(s_3)_{-\frac{16}{3}} \end{bmatrix} = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: & -13 \\ a_2: & -\frac{35}{3} \\ a_3: & -\frac{43}{3} \end{bmatrix} \begin{matrix} \\ \simeq -11.6 \\ \simeq -14.3 \end{matrix}$$

$$= a_2$$

$$\pi'(s_2) = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: & -3 + V(s_2)_{-26/3} \\ a_2: & -6 + V(s_4)_0 \\ a_3: & -3 + V(s_3)_{-\frac{16}{3}} \end{bmatrix} = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: & -\frac{35}{3} \\ a_2: & -6 \\ a_3: & -\frac{25}{3} \end{bmatrix} \begin{matrix} \simeq -11.6 \\ \\ \simeq -8.33 \end{matrix}$$

$$= a_2$$

$$\pi'(s_3) = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: & 0 + V(s_3)_{-\frac{16}{3}} \\ a_2: & -2 + V(s_4)_0 \\ a_3: & 0 + V(s_2)_{-\frac{26}{3}} \end{bmatrix} = \underset{a}{\operatorname{argmax}} \begin{bmatrix} a_1: & -16/3 \\ a_2: & -2 \\ a_3: & -\frac{26}{3} \end{bmatrix}$$

$$= a_2$$

THE NEW POLICY π' ALWAYS CHOOSE a_2, NO MATTER THE CURRENT STATE.

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